

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 17

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • By-catch = $5.9 + 1.9 = 7.8$ t; total = 8.4 t (1) • $7.8 / 8.4 \times 100 = 92.9\%$ (accept 93%) (1)	2
01.2	<i>1 mark.</i> • The nets have a very small mesh size so few non-target animals can escape (1)	1
01.3	<i>1 mark for each point.</i> • They breed at an old age (skate about 11 years; Greenland shark 100-150 years) and produce few young (1) • They may reach a catchable size before they are sexually mature, so fishing can remove the whole breeding population before it has bred (1) • Populations recover slowly, unlike r-selected species (herring, mackerel) which breed young and produce many young; they are caught as by-catch in demersal trawls so fishing continues even though they have no commercial value (1)	3

Question	Marking guidance	Marks
01.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Impacts: population decline of k-selected species, by-catch (immature target fish, over-quota fish, non-commercial species; drift nets - whales, dolphins, turtles; longlines - albatrosses; pair trawls - porpoises), ghost fishing, seabed damage by demersal trawls, coral and seagrass damage, dynamite fishing, food-web effects • Catch quotas: limit weight landed; work well for single-species pelagic shoals but in mixed fisheries over-quota fish are discarded dead so the quota fails to protect them • Gear design: mesh size and square-mesh escape panels let small fish escape (diagonal mesh closes under drag); turtle exclusion devices; acoustic pingers for dolphins; curved hooks, decoys, sinkers and night setting to protect albatrosses; biodegradable / radio-tracked traps to reduce ghost fishing • Restricted effort: limits on boat size, engine power and days at sea; decommissioning; restricted methods - drift-net bans, demersal trawling bans on sensitive seabeds (Rockall Bank, Hatton Bank) • Evaluation: enforcement at sea is difficult; some methods reduce catch or raise costs; quotas need reliable stock data; effectiveness depends on species (single vs mixed shoals) and on international cooperation as fish migrate; as local stocks decline fishing moves to more distant waters • Justified conclusion weighing effectiveness of different methods	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-345; AQA spec 3.5.2.2.	
02.1	<i>1 mark for each point.</i> • Negative feedback responds to any change by re-establishing the previous equilibrium (1) • Only large unusual events overwhelm the feedback so a new equilibrium forms (1) • Example: methane hydrate release at the end of the Permian causing climate change and mass extinction (1)	3

Question	Marking guidance	Marks
02.2	<i>1 mark for each point.</i> • In a diverse system each component is relatively less important so the loss / change of one has a small effect; alternative food sources and relationships buffer the change (1) • Natural: coral reefs and tropical rainforests with complex food webs are stable; simple polar / desert systems fluctuate (1) • Human: agroecosystems focus on a few crops and remove detritivores, pollinators and predators so are vulnerable to pests, disease, weather (1) • Diverse technical systems: a society using a range of renewable energy resources is more secure than one relying on a single source (1)	4
02.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Melting permafrost releasing methane (1) • Reduced albedo as ice / snow melt (1) • Methane hydrate release (1) • Forest and peat fires (1) • Faster soil decomposition (1) • Ocean acidification reducing sequestration (1) • Cirrus cloud formation (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
03.1	<i>1 mark for each point.</i> • Ship rudders that turn in tighter circles / aerogenerator blades that start turning at lower wind speeds (1) • Water-repelling surface of lotus flowers (1) • Coatings for aircraft and ships that reduce fuel consumption / hospital coatings that bacteria cannot stick to (1) • Hooked burrs on seeds that stick to animal fur (1)	4
03.2	<i>1 mark for each point.</i> • The mound absorbs sunlight and warms; hot air inside rises, creating a convection current that draws stale air out of the lower parts (solar-driven ventilation) (1) • Air blowing over the mound has a lower pressure which also draws air out (1) • Copied in shopping complexes / office blocks to give natural ventilation and cooling without air conditioning, so less energy / electricity is used (1)	3
03.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Marsupials: young develop in the pouch so are easier to study than a human foetus, helping understanding of developmental problems (1) • Hippopotamus skin secretes hipposudoric acid (natural sunscreen and antimicrobial) which may improve treatment of burns (1) • Marine sponges produce proteins that prevent rejection of grafts, being developed to prevent rejection of transplanted organs (1)	2
03.4	<i>1 mark.</i> • They are light but strong: hollow with internal struts that prevent bending / breaking, copied in lightweight bridges and roofs (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11-14; AQA spec 3.1.2.1.1.	

Question	Marking guidance	Marks
04.1	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • Rotation: each crop has its own pests; growing a different crop each year (4-5 year cycle) means pests left at the end of a season die before that crop returns, so populations cannot build up early (up to 2) • Beetle banks / hedgerows provide habitat and overwintering sites for natural predators (ground beetles, ladybirds) that eat aphids (up to 2) • Rotation also evens out nutrient demands and includes legumes (up to 2)	4
04.2	<i>1 mark for each point.</i> • Integrated control gives almost the same yield as chemical control (18.1 vs 18.5) with far fewer applications (1-2 vs 6) (1) • It retains most of the biodiversity of cultural control (26 vs 31 species) whereas chemical control alone leaves only 8 (1) • Cultural control alone loses about 12% yield; integrated control combines cultural methods, resistant varieties and carefully timed specific non-persistent pesticides only when essential - best balance of yield, cost and impact (1)	3
04.3	<i>1 mark for each point.</i> • Large numbers of males are sterilised with gamma radiation and released (1) • Females of many insects mate only once and store sperm, so mating with a sterile male produces no offspring; enough sterile males reduce fertile matings below replacement level (screw worm fly, Mediterranean fruit fly) (1) • Only the target species' females mate with them so non-target species are unaffected; fails if sterilised males behave abnormally or re-colonisation occurs (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 314-318; AQA spec 3.5.1.1.	
05.1	<i>Correct answer only.</i> • Environmental DNA (eDNA) analysis (1)	1
05.2	<i>1 mark for each point.</i> • Transmitters are too heavy for small birds / their batteries would not last for a long-term study (1) • Geolocator tags are small and only record time and light levels; latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded (1)	2
05.3	<i>1 mark for each point.</i> • Each separate (regional) population has its own unique gene pool (1) • DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	

Question	Marking guidance	Marks
06.1	<i>1 mark for each point.</i> • Diversity: HEP (wet winters), solar (summer) and wind peak at different times so supply is more reliable - diverse systems resist change (1) • No single source could meet demand all year; battery storage on the island grid smooths fluctuations (1) • Calm, dry, dull periods still leave gaps so diesel generators provide security (fossil fuel use minimised, not yet eliminated) (1)	3
06.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Community ownership (Isle of Eigg Trust with Scottish Wildlife Trust) managing the island for people and wildlife (1) • Wood biomass heating; insulation such as straw-bale housing; some electric vehicles (1) • Habitat protection and creation: fencing rare plants from grazing, planting indigenous woodland (rowan, birch, willow); eagles and hen harriers nesting (1) • Recyclables shipped off; garden / food waste composted; low-impact food production; beach litter collection (1)	3
06.3	<i>1 mark for each point.</i> • Small groups can organise, agree and act quickly; decisions benefit people who see the results directly (1) • Resources and wastes are local and visible so feedback is immediate, unlike cities supplied from far away (1)	2
06.4	<i>1 mark for each point.</i> • Renewable energy is abundant but geographically and temporally variable; a continental grid moves electricity from areas of surplus (desert solar, offshore wind) to areas of demand (1) • DC grids do not need every generator synchronised in phase as AC grids do, so coordination across many countries is easier (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 384-389; AQA spec 3.6.4.	
07.1	<i>1 mark for each point.</i> • Natural processes are driven by renewable, low energy-density solar energy captured by photosynthesis; human processes usually use high energy-density non-renewable fossil fuels (1) • Natural processes occur at low temperatures because enzymes lower the activation energy of reactions; human processes need high temperatures and pressures (1) • Natural processes produce no toxic wastes / their products are raw materials for other processes; industrial processes emit CO₂ and NO_x (1) • So natural fixation is sustainable indefinitely with a low carbon footprint (1)	4
07.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Hydrological cycle (evaporation, precipitation) (1) • Carbon cycle / photosynthesis (1) • Atmospheric circulation / winds (1) • Thermohaline circulation (density differences from heating / evaporation) (1) • Decomposition (1)	3

Question	Marking guidance	Marks
07.3	<i>1 mark for each point.</i> • Bioleaching and phytomining extract metals at low temperature instead of high-temperature smelting (1) • Anaerobic digestion and composting treat wastes; enzymes in industry (biological washing powders, biofuels from algae / microorganisms) (1) • Heat pumps and solar thermal capture low-density heat; low-temperature manufacturing using biological polymers; fuel cells instead of combustion (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
08.1	<i>1 mark for each point.</i> • SSSI: UK legislation (Wildlife and Countryside Act); best examples of UK flora, fauna, geological features; management plan agreed with owner who must notify Operations Likely to Damage (ploughing, drainage, pesticides, tree planting / removal) (1) • SPA: EU Birds Directive; places of international importance for birds (with SACs under the Habitats Directive form the Natura 2000 network; may restrict activities in surrounding areas) (1) • Ramsar: Ramsar Convention (intergovernmental agreement); internationally important wetlands, originally for waterfowl, now wetland conservation and wise use generally (1) • MCZ: UK marine legislation (Marine and Coastal Access Act) under the W&CA framework; protects marine habitats / species by restricting activities such as fishing and dredging; NNR: best SSSIs with complete communities (1)	4
08.2	<i>1 mark for each point.</i> • Legal status: restrictions on damaging activities within the area (and sometimes in surrounding areas, eg drainage) (1) • Management agreements between owners and statutory bodies to maintain suitable conditions (1) • But it does not protect against external influences such as air pollution or climate change; very few sites are state-owned so protection depends on owner cooperation (1)	3
08.3	<i>1 mark for each point.</i> • Much of the UK's wildlife lives in habitats created by farming (hay meadows, hedgerows) which only survive if the farming continues (1) • Managing land for wildlife costs money / reduces income so farmers need payments (points for beetle banks, hedgerows, buffer strips, wild bird seed, low-input grassland) (1) • Benefits everyone so society pays through central funding; legislation cannot compel positive management (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30, 38-39; AQA spec 3.1.2.3.2.	
09.1	<i>Correct answer only.</i> • Its mass (1)	1

Question	Marking guidance	Marks
09.2	1 mark for each point. - The 24-hour rotation period means the surface is not exposed to sunlight for long, so temperature extremes / excessive heating and cooling are reduced (1) - The axis is tilted relative to the orbit so the amount of insolation at each latitude changes through the year, producing seasonal variations in conditions (1)	2
09.3	1 mark for each point. - The distance from the Sun gives surface temperatures between 0 and 35 °C in most areas: too close and water would boil / evaporate, too far and it would freeze (1) - Atmospheric pressure (produced by the mass of gases held by gravity) is high enough to stop the water boiling / evaporating away; at much lower pressure there would be no liquid water (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Protected areas: Korup NP Cameroon (1260 km², 600 tree species, endemic <i>Ancistrocladus korupensis</i> with anti-HIV / antimalarial alkaloids); Tumucumaque NP Brazil (39 000 km², adjoining French Guiana reserve); Alto Maues; Rio Bravo Belize funded by charities, companies and ecotourism - Debt for nature swaps; sustainable exploitation: selective logging of high-value timber (mahogany) funding conservation; ITTO aims (largely unmet); CITES listing of timber species (Honduras mahogany Appendix II) - Ecotourism income (but visitor damage in Costa Rica, Malaysia); community projects and employment; carbon sequestration payments - Effectiveness: protected areas work where enforced but face illegal logging, mining, poaching, roads, conflict; drivers (population growth, demand for beef, soya, palm oil, timber, minerals) originate outside the forest; climate change cannot be controlled locally - Judgement: combination of large protected areas, economic alternatives for local people and reduced international demand needed; overall loss continues	9
10.2	1 mark. Carbon sequestration reducing climate change; genetic resources / medicines; rainfall regulation; global biodiversity (1)	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 56-57; AQA spec 3.1.2.3.4.	
11.1	Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Marine productivity: light, nutrients, upwelling; where fisheries occur • Methods: demersal / pelagic trawling, purse seining, drift nets, long lines, shellfish traps - effectiveness, selectivity, energy, impacts • Impacts: overfishing (r vs k selection, catchable vs breeding size), changed age structure, by-catch (albatross, porpoise, turtles, shrimp), ghost fishing, seabed / reef / seagrass damage, dynamite, food web effects (sandeels, tritons) • Population dynamics and MSY: Russell formula, difficulties of data, indicators from catches (CPUE, mean size, age), research on eggs and larvae • Management: quotas, mesh design, escape panels, TEDs, pingers, hook design, effort limits, method bans, NTZs, closed seasons, size limits, protected individuals, population seeding, biodegradable gear; EU CFP, IWC, CCAMLR • Evaluation with examples (cod, orange roughy, herring recovery, NTZ spillover); role of aquaculture; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-351; AQA spec 3.5.2.	25

Question	Marking guidance	Marks
11.2	Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Physical features: mass / gravity retaining an atmosphere and giving pressure for liquid water; distance from the Sun and the Goldilocks zone; speed of rotation reducing temperature extremes; tilt producing seasons; magnetic field deflecting the solar wind; the Moon stabilising the orbit and producing tides • Importance of liquid water: solvent, transport, temperature control, anomalous expansion, high specific heat capacity, habitats, absorption of UV before the ozone layer • Development of life: Archaea at vents; photosynthesis; oxygen reacting with iron then accumulating; ozone layer allowing colonisation of land; aerobic organisms • Life changing conditions: carbon sequestration offsetting solar brightening, biogeochemical cycles, transpiration and rainfall, regulation of atmospheric gases • Methods of studying past conditions: proxy data, isotope ratios, ice cores, Mg:Ca ratios; understanding of continental drift and ocean currents • Links to current threats (climate change, ozone depletion) and a justified conclusion about the interdependence of physical and biological factors Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-9; AQA spec 3.1.1.	25